University of Florida, Department of Mathematics

Numerical Linear Algebra First-Year Exam, January 2015

Do 4 (four) problems.

1. Assume $A \in \mathbb{R}^{m,n}$ with $m \geq n$, $\text{rank}(A) = n$ and $b \in \mathbb{R}^m$.

(a) Define the least squares solution to $Ax = b$.

(b) Derive the normal equations for the least squares problem.

(c) Prove that $A^T A$ is invertible.

(d) Prove that the unique solution to the least squares problem is $(A^T A)^{-1} A^T b$.

(e) Describe how to solve the least squares problem using the QR decomposition of A.

2. This problem has two parts.

(a) Compute $\det(\lambda I + uv^*)$ when $\lambda \in \mathbb{C}$, I is the $m \times m$ identity matrix, and $u, v \in \mathbb{C}^m$.

(b) Prove necessary and sufficient conditions for $I + uv^*$ to be nonsingular and when it is, give a formula for its inverse.

3. This problem has three parts.

(a) If P is a projector, prove that $\text{null}(P) \cap \text{range}(P) = \{0\}$ and $\text{null}(P) = \text{range}(I - P)$.

(b) Prove that P is an orthogonal projector if and only if it is Hermitian.

(c) If $q_1, \dots, q_n$ is an orthonormal basis for the subspace $V \subset \mathbb{C}^m$ with $m > n$, prove that the orthogonal projector onto V is QQ^*, where Q is the matrix whose columns are the q_j.

4. This problem has three parts.

(a) Prove that $\|A\|_F = \text{trace}(A^* A)^{1/2}$.

(b) Prove that $\|A\|_F = (\sum \sigma_i^2)^{1/2}$, where $\{\sigma_i\}$ are the singular values of A counted with multiplicity.

(c) If both A and U are in $\mathbb{C}^{m,m}$ and U is unitary, prove that $\|UA\|_F = \|AU\|_F = \|A\|_F$.

5. Define a normal matrix and prove that the following are equivalent.

(a) A is normal.

(b) A is unitarily diagonalizable.

(c) $\|A\|_F = (\sum |\lambda_i|^2)^{1/2}$, where $\{\lambda_i\}$ are the eigenvalues of A counted with multiplicity.